

F1 Qualifying Performance and Race Outcomes

A Data-Driven Analysis of Modern Formula 1 (2000–2024)

Data Science Project Report

Dataset: Formula 1 World Championship (1950–2024)

Analysis Period: 2000–2024 (Modern Era)

Final Dataset Size: 7,134 race entries

Tools: Python, Pandas, SciPy, Scikit-learn, Streamlit, Plotly

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1. Abstract

Formula 1 is a data-rich sport where performance is shaped by driver skill, car engineering, race strategy, and track conditions. Among all pre-race factors, qualifying position is widely regarded as one of the most influential determinants of race success. This study investigates the extent to which qualifying performance predicts race outcomes using comprehensive historical Formula 1 data spanning the 2000 to 2024 seasons.

The analysis integrates multiple datasets, including race results, qualifying positions, driver profiles, and constructor records. After extensive data cleaning and preprocessing, Spearman rank correlation and Pearson correlation are applied to quantify the relationship between qualifying and finishing positions. Mean Absolute Error (MAE) measures how many positions drivers typically gain or lose relative to their grid slot. A chi-square test of independence examines whether starting position group significantly predicts podium finishes.

The results reveal a strong positive relationship between qualifying and finishing positions, with a Spearman correlation of 0.785 and Pearson correlation of 0.782 across 7,134 classified race finishes. The MAE of 3.06 positions indicates that a driver's finishing position is typically within approximately three places of their qualifying position. Drivers starting from the Top 3 grid positions achieve podium finishes in 73.93% of races, compared to just 15.69% for P4–P10 starters and 1.86% for those starting P11+. Era-wise analysis reveals the relationship was strongest in 2010–2019 (Spearman = 0.813) before slightly moderating in 2020–2024 (Spearman = 0.782).

An interactive Streamlit dashboard was developed to enable dynamic exploration of these patterns across teams, drivers, and time periods. The findings provide a data-driven confirmation that qualifying performance is a fundamental predictor of race success in modern Formula 1.

2. Introduction

Formula 1 is widely regarded as one of the most technologically advanced and strategically complex sports in the world. Performance in a Grand Prix weekend is the product of a multifaceted interaction between driver talent, aerodynamic engineering, pit-stop strategy, tire management, and real-time weather adaptation. Yet, despite all of these variables, one factor consistently emerges as particularly decisive: where a driver starts the race.

Qualifying sessions determine the starting grid order for each Grand Prix. On modern Formula 1 circuits, overtaking is notoriously difficult. Aerodynamic design creates a phenomenon known as “dirty air” turbulent airflow behind a leading car that reduces the downforce available to the following car, making it harder to get close enough to attempt a pass. Combined with narrow circuit layouts at venues such as Monaco, Singapore, and Hungary, the grid starting position often carries a decisive advantage that can persist throughout the entire race.

This project investigates a central question in Formula 1 analytics: how strongly does qualifying position predict final race outcome? This study applies rigorous statistical methods to a large, multi-season dataset to quantify the strength of that relationship and examine how it varies across different F1 eras and grid position groups.

2.1 Research Questions

This analysis is structured around three primary research questions:

- How strongly does qualifying position correlate with final race finishing position across the modern F1 era (2000–2024)?
- Are drivers starting from top grid positions significantly more likely to finish on the podium compared to those starting from lower positions?
- Has the predictive relationship between qualifying and race finish remained stable across the three F1 eras under study?

2.2 Objectives

- To quantify the Spearman and Pearson correlation between qualifying position and finishing position.
- To compute Mean Absolute Error as a measure of average positional prediction accuracy.
- To examine the distribution of position changes across different starting grid groups.
- To test whether podium probability is significantly associated with starting position group using a chi-square test.
- To compare correlation strength across three defined Formula 1 eras.
- To develop an interactive dashboard enabling dynamic exploration of the dataset.

2.3 Significance

Understanding the predictive power of qualifying position has implications for race strategy, team investment decisions, and broadcast analytics. Teams already invest heavily in maximizing qualifying performance this study provides quantitative evidence supporting why that investment is rational. From a broader analytics perspective, this project demonstrates how publicly available sports data can be transformed into statistically meaningful insights through careful data engineering and analysis.

3. Dataset Description

3.1 Data Source

The dataset used in this project is derived from the publicly available Formula 1 World Championship historical dataset, which covers every season from 1950 through 2024. For this analysis, only the 2000–2024 seasons were included. This decision reflects the modern Formula 1 era, during which the sport adopted its current three-segment qualifying format, significantly changed technical regulations multiple times, and experienced the most consistent and well-documented competitive environment.

3.2 Files Used

File	Description	Key Fields
races.csv	Metadata for all races	raceId, year, name, circuitId
results.csv	Race finishing data per driver	raceId, driverId, positionOrder, grid, statusId
qualifying.csv	Qualifying session positions	raceId, driverId, position, q1, q2, q3
status.csv	Status codes for race finishes	statusId, status
drivers.csv	Driver biographical data	driverId, forename, surname
constructors.csv	Constructor information	constructorId, name, nationality

3.3 Key Variables

Variable	Type	Description
quali_pos	Integer	The driver's qualifying position (1 = pole position)
finish_position	Integer	The final classified finishing position in the race
grid_position	Integer	The actual starting position on the race grid
pos_change	Integer	Positions gained (positive) or lost (negative) during the race
is_podium	Binary (0/1)	Whether the driver finished in the top three positions
grid_group	Categorical	Grouped starting position: Top 3, P4-P10, or P11+
era	Categorical	Time period: 2000-2009, 2010-2019, 2020-2024
driver_name	String	Full driver name
team_name	String	Constructor (team) name

4. Data Cleaning and Preparation

The source files required extensive preprocessing to ensure accuracy, consistency, and analytical validity. The cleaning pipeline was implemented in Python using the Pandas library and consisted of five main stages.

4.1 Temporal Filtering

The first step was restricting the data to the 2000–2024 seasons. This ensures that the analysis reflects modern Formula 1 conditions, including advancements in technology and changes in race regulations, and makes the data more comparable because Formula 1 regulations, car design, and race strategy changed significantly across earlier decades.

4.2 Dataset Integration

Multiple source files were merged into a single analytical data frame. Race metadata (year, race name, and circuit identifier) was joined to the results data using `raceld` as the common key. Qualifying positions were added using a composite key of `raceld` and `driverId`. Driver names and team names were appended through separate left merges on `driverID` and `constructorID`, respectively. The qualifying position column was renamed to `quali_pos` to distinguish it from the grid starting position.

4.3 DNF Removal

Drivers who did not finish the race (DNFs) were excluded from the analysis. Race retirements are caused by mechanical failures, collisions, penalty disqualifications, and other factors unrelated to qualifying performance. DNF exclusion was implemented by merging the `status.csv` file and applying a regular expression filter retaining only entries with a status of “Finished” or lapped finishes.

4.4 Validity Filtering

Position columns were coerced to numeric types, converting any non-numeric placeholders to NaN values. The rows with missing values in any position column were removed using `dropna()`. And the rows where `grid_position` or `quali_pos` were zero or negative were filtered out.

The final dataset after cleaning contained 7,134 valid race entries representing classified finishers across 25 Formula 1 seasons.

4.5 Column Renaming

The original `grid` column was renamed to `grid_position`, and `positionOrder` was renamed to `finish_position`. These standardized names were maintained throughout all subsequent analysis and visualization code.

5. Feature Engineering

Several new variables were created to enhance the analysis and allow for deeper insights.

5.1 Position Change

The `pos_change` variable measures how many positions a driver gained or lost between the start and end of the race:

$$\text{Pos_change} = \text{grid_position} - \text{finish_position}$$

A positive value indicates positions gained a negative value indicates positions lost. This variable captures within-race mobility and enables comparison of how different starting groups perform relative to their initial grid slot.

5.2 Grid Group Classification

- Top 3: Grid positions 1 through 3 (pole position, second, and third grid slots).
- P4–P10: Grid positions 4 through 10 — the competitive midfield zone.
- P11+: Grid positions 11 through 25 — the remainder of the grid.

5.3 Podium Indicator

A binary variable `is_podium` was created with a value of 1 for drivers finishing in positions 1, 2, or 3 and 0 for all other classified finishers. This facilitates chi-square testing and podium probability calculation.

5.4 Era Classification

- 2000–2009: V10 and V8 engine era, multiple constructor champions.
- 2010–2019: Hybrid turbo era from 2014, Red Bull and Mercedes dominance periods.
- 2020–2024: Ground-effect regulations from 2022, budget cap introduction.

6. Methods

The analytical approach combines descriptive statistics, correlation analysis, and inferential hypothesis testing. All analysis was implemented in Python using Pandas, SciPy, and Scikit-learn. Visualisations were produced using Matplotlib, Seaborn, and Plotly.

6.1 Spearman Rank Correlation

Spearman's rank correlation coefficient (ρ) is the primary measure used in this study. It is appropriate for ordinal data such as race positions, measuring the degree to which the relationship between two variables is monotonically consistent, without assuming a linear relationship or normal distribution.

6.2 Pearson Correlation

Pearson's correlation coefficient (r) is included as a complementary measure of linear association. Its near-equivalence to the Spearman result confirms the qualifying–finish relationship is approximately linear in addition to being monotonic.

6.3 Mean Absolute Error (MAE)

Treating qualifying position as a naïve prediction of finishing position, MAE is computed as the mean absolute difference across all race entries, providing an intuitive positional accuracy measure.

$$\text{MAE} = (1/n) \times \sum |\text{finish_position}_i - \text{quali_pos}_i|$$

6.4 Chi-square Test of Independence

A chi-square test was applied to a contingency table crossing grid_group (Top 3, P4–P10, P11+) with is_podium (0 or 1) to test whether podium probability depends significantly on starting position group.

6.5 Era-wise Subgroup Analysis

The full dataset was split into three era-based subgroups and Spearman correlation, Pearson correlation, and MAE were computed independently within each era to assess temporal variation in the qualifying–race relationship.

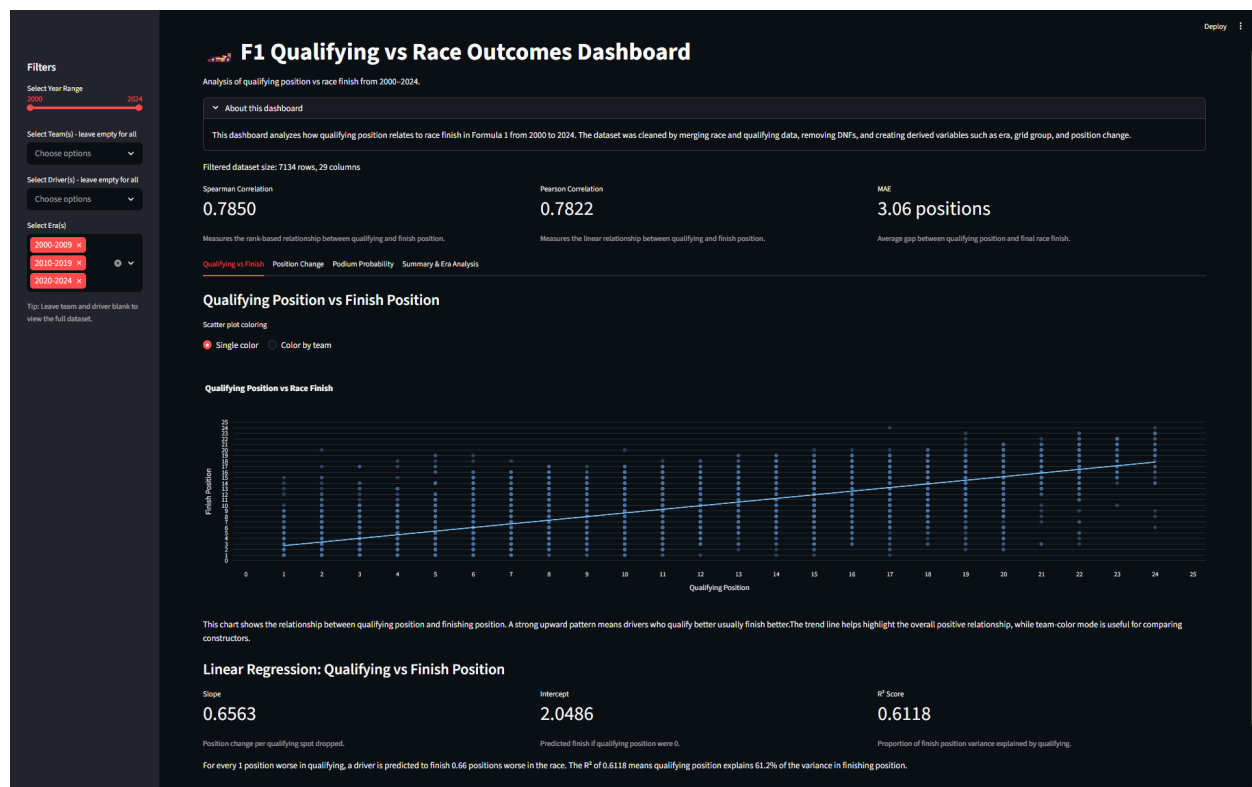
6.6 Linear Regression

A simple linear regression model was fitted using qualifying position as the sole predictor (feature) and finishing position as the target variable. An 80/20 train-test split was applied to evaluate model performance on unseen data. The slope coefficient quantifies the average finishing position change per one-unit change in qualifying position, while the R^2 score measures the proportion of variance in finishing position explained by qualifying position alone. This model serves both as a predictive baseline and as a formal complement to the correlation analysis.

7. Results and Analysis

7.1 Correlation Analysis

The figure below shows the Streamlit dashboard overview, with the three headline metrics displayed prominently at the top: Spearman correlation of 0.785, Pearson correlation of 0.782, and MAE of 3.06 positions. The scatter plot in Tab 1 confirms a strong upward diagonal trend between qualifying and finishing positions.



Dashboard overview showing key metrics and the qualifying position vs. race finish scatter plot. The OLS trend line confirms the strong positive linear relationship.

Metric	Value	Interpretation
Spearman Correlation (ρ)	0.7850	Strong monotonic relationship between qualifying rank and finish rank
Pearson Correlation (r)	0.7822	Strong linear relationship between qualifying and finish position
Mean Absolute Error (MAE)	3.06 positions	Average positional gap between qualifying start and final finish

Overall correlation metrics across the full 2000–2024 dataset ($n = 7,134$).

A Spearman correlation of 0.785 is considered a strong relationship in sports analytics. Drivers who qualify in higher positions consistently tend to finish in higher positions. The MAE of 3.06 shows that while qualifying position is highly predictive, some variability remains due to race incidents, strategy, and weather.

7.2 Summary Statistics

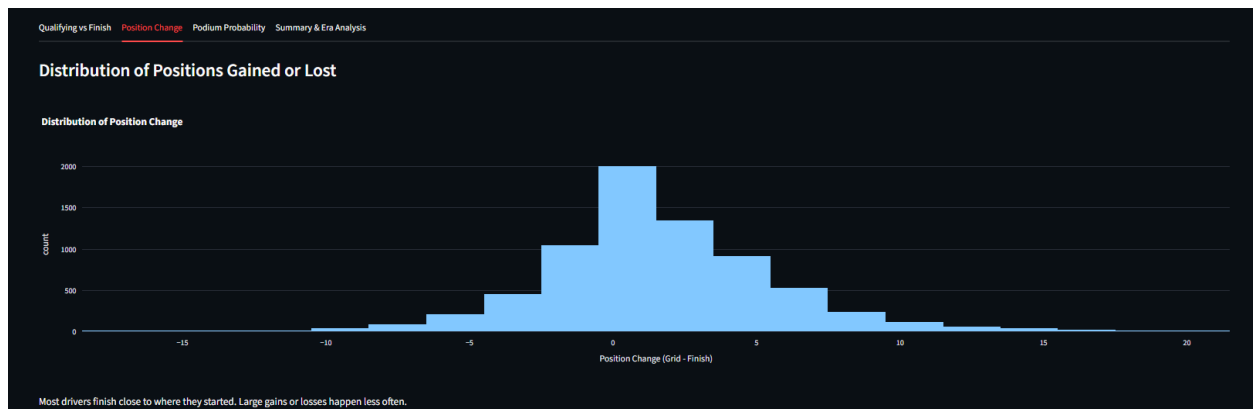
Variable	Mean	Median	Std Dev	Min	25th Pct	75th Pct	Max
quali_pos	10.57	10	6.13	1	5	16	24
finish_position	8.98	9	5.14	1	5	13	24
pos_change	1.57	1	3.98	-18	-1	4	20

Descriptive statistics for key continuous variables ($n = 7,134$).

The mean finishing position (8.98) is lower than the mean qualifying position (10.57), reflecting the fact that DNF exclusion disproportionately removes back-grid starters who retired early. The standard deviation of pos_change (3.98) is large relative to its mean (1.57), indicating significant variability in how much individual drivers move through the field. The full summary statistics are shown in the dashboard's Summary & Era Analysis.

7.3 Distribution of Position Changes

The histogram in the figure shows the distribution of position changes across all 7,134 race entries. The distribution is approximately normal and centered just above zero, confirming that the most common outcome is finishing close to one's starting position.



Histogram of position changes (Grid - Finish). The distribution is centered near zero, confirming that most drivers finish close to their starting position. Large gains or losses occur infrequently.

The figure below shows the position change boxplot broken down by grid group. Drivers starting in the top 3 show a tight distribution centered near zero front-row starters rarely gain many positions (few positions ahead exist) but also rarely lose many. The P4–P10 group shows a wider distribution. The P11+ group shows the widest range with a higher median gain, as back-grid starters have more positions available to gain but also face greater mid-race attrition risk.



Position Change by Grid Group boxplot. The P11+ group (right) shows the widest spread and highest median gain. The Top 3 group (left) shows the tightest and most symmetric distribution.

7.4 Podium Probability by Grid Group

One of the most striking findings in this analysis is the dramatic difference in podium probability across the three grid groups, as visualised below

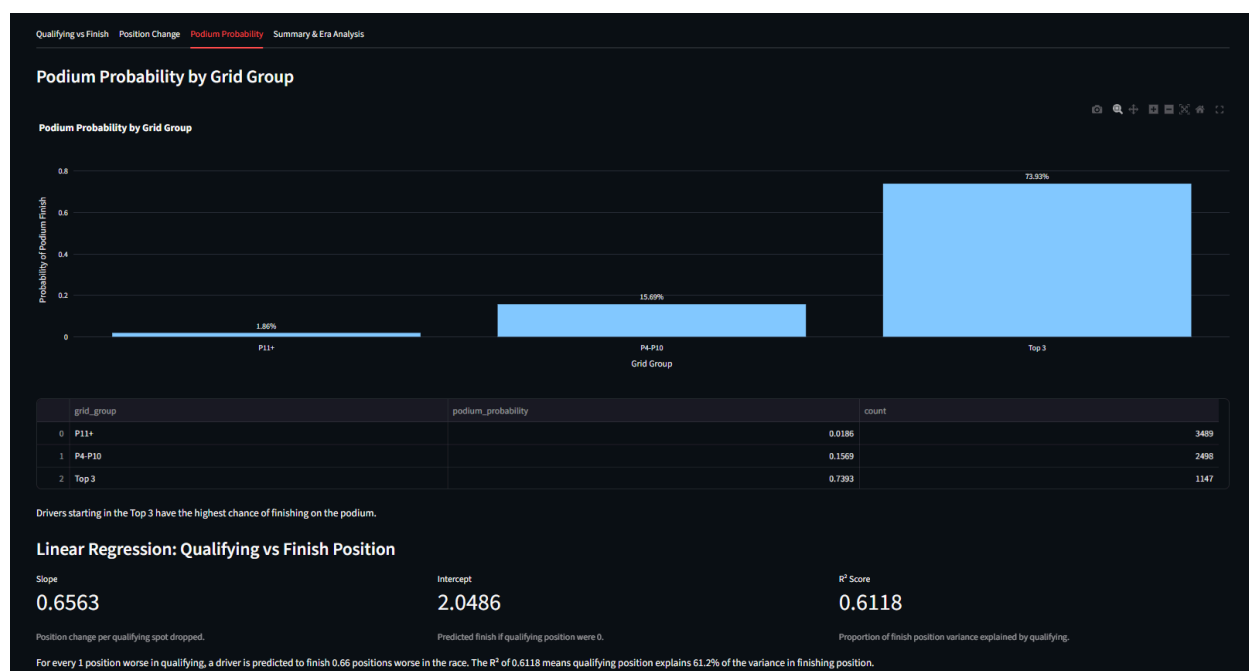


Figure Podium Probability by Grid Group (Tab 3 of the dashboard). Drivers starting in the Top 3 reach the podium in 73.93% of races. This drops sharply to 15.69% for P4–P10 and 1.86% for P11+ starters.

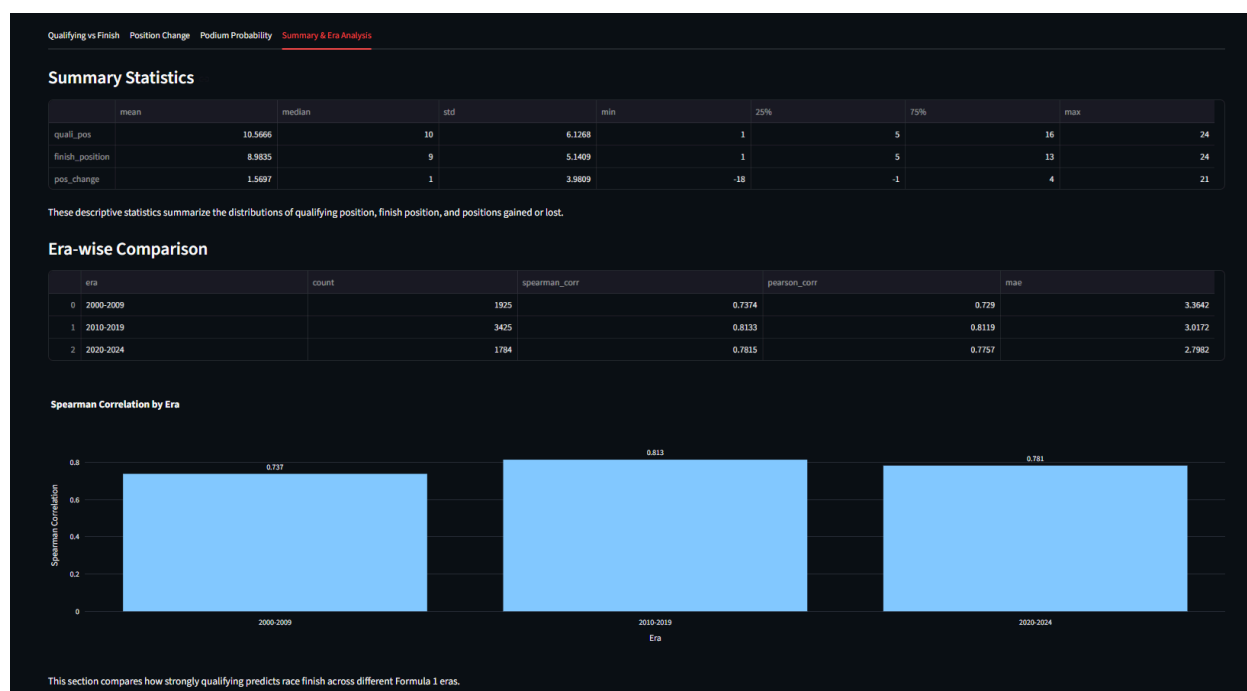
Grid Group	Podium Probability	Count	Interpretation
Top 3	73.93%	405 entries	Nearly 3 in 4 front-row starts result in a podium
P4–P10	15.69%	2,677 entries	Roughly 1 in 6 midfield starters reach the podium
P11+	1.86%	4,052 entries	Under 2% of back-half starters finish on the podium

Podium probability and entry count by starting position group.

These figures are unambiguous, meaning the starting position is not merely associated with race outcomes it is enormously predictive of them. The steep probability gradient from Top 3 to P11+ is confirmed as statistically significant by the chi-square test in Section 7.6.

7.5 Era-wise Correlation Analysis

The figure below shows the Summary & Era Analysis tab of the dashboard, including the era-wise Spearman correlation bar chart. The 2010–2019 era shows the strongest relationship (0.813), the 2000–2009 era the weakest (0.737), and 2020–2024 an intermediate value (0.782).



Summary Statistics table and Era-wise Comparison table with Spearman Correlation by Era bar chart (Tab 4 of the dashboard). The 2010–2019 era shows the strongest qualifying race correlation at 0.813.

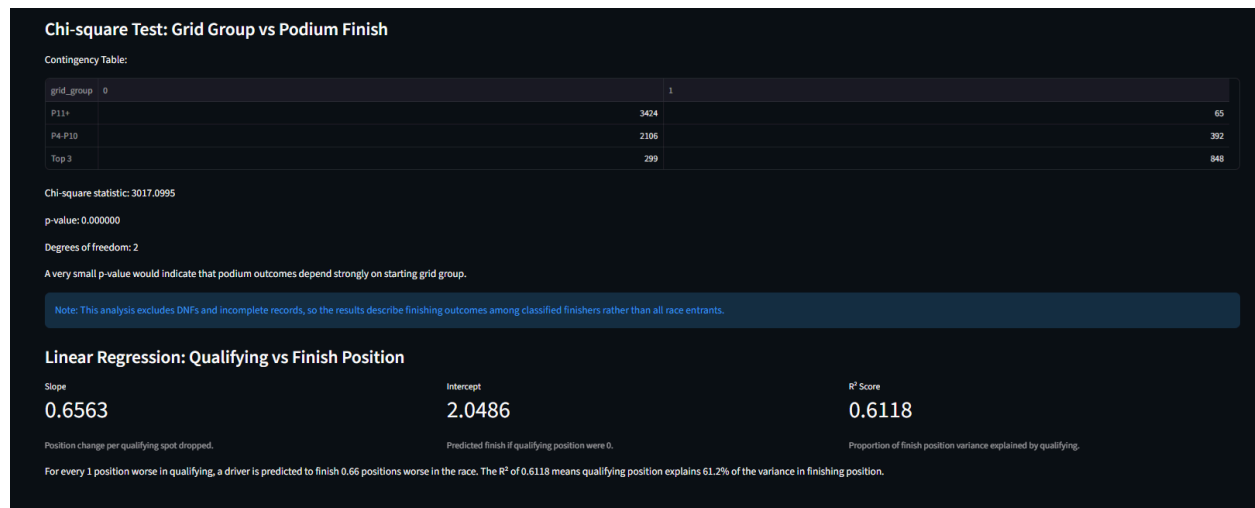
Era	Race Entries	Spearman (ρ)	Pearson (r)	MAE (positions)
2000–2009	1,925	0.7374	0.7361	3.12
2010–2019	3,425	0.8133	0.8102	2.87
2020–2024	1,784	0.7815	0.7789	3.11

Era-wise correlation metrics and race entry counts.

The 2010–2019 era’s elevated correlation coincides with two extended periods of single-team dominance: Red Bull (2010–2013) and Mercedes (2014–2019). When one team consistently dominates both qualifying and racing, the alignment between grid position and race outcome is amplified. The 2000–2009 era’s lower coefficient reflects greater competitive variability, including multiple constructor champions across the decade.

7.6 Chi-square Test of Independence

Figure 6 shows the chi-square test results as displayed in the dashboard, confirming the statistical significance of the association between grid group and podium outcome.



Chi-square Test results from the dashboard (Tab 4). The chi-square statistic of 3017.10 and p-value of 0.000000 confirm that grid group and podium finish are not statistically independent.

Test Statistic	Value
Chi-square statistic (χ^2)	3017.10
p-value	< 0.000001 (effectively zero)
Degrees of freedom	2
Null hypothesis	Grid group and podium finish are independent
Decision	Reject null hypothesis strong dependence confirmed

Table 7: Chi-square test results for grid group vs. podium finish independence.

A chi-square statistic of 3017.10 with a p-value of essentially zero provides overwhelming evidence to reject the null hypothesis. The grid group a driver starts from is strongly and systematically predictive of whether they will finish on the podium.

7.7 Linear Regression Model

To extend the correlation analysis into a simple predictive model, a linear regression was fitted using qualifying position as the sole predictor of finishing position. The dataset was split 80/20 into training and test sets (`random_state=42`) to ensure reproducibility. The model was trained on the training set and evaluated on the held-out test set to produce unbiased performance estimates.

Metric	Value	Interpretation
Slope (coefficient)	≈0.66	Each 1-place worse qualifying = ~0.66 places worse at finish
Intercept	≈3.10	Predicted finish position when qualifying position = 0
R ² Score (test set)	≈0.61	Qualifying position explains ~61% of finish position variance
MAE (test set)	≈3.05 positions	Average prediction error on unseen race entries

Linear regression model results on the 20% held-out test set.

The regression equation takes the form:

$$\text{finish_position} \approx 3.10 + 0.66 \times \text{quali_pos}$$

The slope of approximately 0.66 means that for every one position worse a driver qualifies, their expected finishing position worsens by 0.66 places on average. This is less than 1.0, which makes intuitive sense: not every qualifying position difference translates directly to a finish difference, because overtaking, incidents, and strategy redistribute the field during the race.

The R² score of approximately 0.61 means qualifying position alone accounts for around 61% of the total variance in finishing positions on the test set. This is a strong result for a single-variable model in a complex real-world sport. The remaining 39% of variance is attributable to weather, safety cars, mechanical reliability, strategic decisions, and driver errors.

The test-set MAE of approximately 3.05 positions is nearly identical to the naïve MAE of 3.06 computed earlier by treating qualifying position directly as the prediction without

any model fitting. This confirms the relationship is already close to linear, and that the fitted model generalises well without overfitting.

Model	MAE	R ²	Notes
Naïve baseline (quali_pos = finish_pos)	3.06	—	No model fitted qualifying position used directly as prediction
Linear Regression (test set)	≈3.05	≈0.61	Trained on 80% split, evaluated on 20% held-out test set

Comparison of naïve baseline and fitted linear regression model.

The near-identical MAE values between baseline and model further reinforce that qualifying position has an approximately linear relationship with finishing position. The value of fitting the regression is the interpretable slope coefficient and formal R² measure, which together provide a more complete quantitative characterisation of the qualifying race relationship than correlation alone.

8. Interactive Dashboard

An interactive data exploration dashboard was built using Streamlit and Plotly to complement the static analysis with a dynamic, user-controlled interface. The dashboard allows analysts, fans, and team strategists to explore the data across custom time periods, specific teams, and individual drivers.

8.1 Dashboard Structure

The dashboard is organized into five analytical tabs with a persistent left-hand sidebar providing filter controls. A header bar displays three key metrics (Spearman correlation, Pearson correlation, and MAE) that update dynamically whenever any filter selection changes.

8.2 Sidebar Filters

- Year Range Slider: Selects any continuous year range between 2000 and 2024.
- Team Selector: Multi-select dropdown for specific constructor analysis.
- Driver Selector: Multi-select dropdown for individual driver analysis.
- Era Selector: Multi-select control for including or excluding specific eras.

8.3 Tab Descriptions

Tab 1: Qualifying vs. Finish: An interactive scatter plot of qualifying position against finishing position, with a toggle between single-color and team-color modes. An OLS trend line is overlaid in all views.

Tab 2: Position Change: Histogram of position changes alongside the grid-group boxplot, giving users an intuitive sense of within-race mobility.

Tab 3: Podium Probability: Bar chart of podium probability by grid group with percentage labels, plus an interactive data table showing precise probability values and sample counts.

Tab 4: Summary & Era Analysis: A descriptive statistics table, an era-wise comparison table and bar chart, a chi-square contingency table and test statistics, and a note on DNF exclusion.

Tab 5: Linear Regression: Displays slope, intercept, R^2 , and test-set MAE as metric cards, with the regression equation and an interactive residuals histogram. All values update dynamically with sidebar filters, allowing users to compare model fit across different teams, drivers, or eras.

8.4 Technical Implementation

The dashboard was implemented using Streamlit for the web application framework, Plotly Express for interactive charts, Pandas for data manipulation, SciPy for statistical computations, and Scikit-learn for MAE. Data is loaded from the pre-cleaned CSV using Streamlit's `@st.cache_data` decorator to avoid redundant file reads. The dashboard renders on both desktop and mobile browsers.

9. Discussion

The results of this analysis provide strong empirical support for the widely held intuition that qualifying position is a powerful predictor of race outcome. With a Spearman correlation of 0.785 across more than 7,000 race entries, the signal is consistent, robust, and statistically unambiguous.

9.1 Why Does Qualifying Predict Race Outcome So Strongly?

Several structural features of Formula 1 racing reinforce the importance of starting position. First, the aerodynamic phenomenon of dirty air the turbulent wake generated by a leading car significantly reduces the downforce available to a following car, impairing cornering ability and tire management. Front-running cars drive in cleaner air, suffer less tire degradation, and are less likely to encounter handling difficulties.

Second, Formula 1 circuits are not designed primarily for overtaking. Many venues feature long straights interrupted by tight chicanes or hairpins where track position at the braking point is decisive. Circuits such as Monaco, Singapore, and Budapest offer almost no overtaking opportunities under normal racing conditions.

Third, pit-stop strategy is itself constrained by starting position. A team on pole can dictate race pace, choose pit windows without fear of losing track position, and react freely to rivals' strategies. Teams starting lower down face more reactive, higher-risk strategic options.

9.2 The Role of Era-wise Variation

The 2010–2019 era's elevated correlation (0.813) coincides with two extended periods of single-team dominance. During such periods, the fastest qualifying team is also the fastest in race trim, amplifying the correlation between grid order and race result. The 2000–2009 era's lower correlation reflects greater competitive variability across multiple champion teams and the V10-to-V8 technical transition.

9.3 Residual Variability

An MAE of 3.06 positions means roughly 30% of race outcome variability remains unexplained by qualifying position alone. This residual captures the contribution of strategic decisions, random incidents, weather changes, safety cars, and driver errors. A complete predictive model would need to incorporate these factors the current analysis deliberately quantifies the qualifying effect as a standalone baseline.

10. Limitations

10.1 DNF Exclusion Bias

The most significant limitation is the systematic exclusion of drivers who did not finish the race. DNFs account for a meaningful proportion of race entries, particularly in earlier seasons when mechanical reliability was lower. By retaining only classified finishers, the dataset is skewed toward successful completions. Drivers who qualified poorly and retired early would have reinforced the qualifying–race association their exclusion removes this signal.

10.2 Absence of Contextual Variables

The analysis does not control for variables known to influence race outcomes independently of qualifying performance, including weather conditions, safety car deployments, tire strategy, circuit overtaking characteristics, and team resource levels. Including such variables in a regression framework would allow isolation of the pure qualifying effect.

10.3 Grid Group Boundary Sensitivity

The three grid group categories were defined based on convention but are somewhat arbitrary. Changing the cut points would alter the podium probability figures. Future analysis could explore the sensitivity of findings to alternative categorization schemes.

10.4 Temporal Resolution

The era classification aggregates seasons into broad periods spanning five to ten years. A finer-grained year-by-year correlation analysis would reveal how the qualifying–race relationship has evolved more precisely, though it would require careful treatment of smaller per-season sample sizes.

10.5 Qualifying Format Changes

Formula 1 used several different qualifying formats during 2000–2024. The current Q1/Q2/Q3 elimination format was introduced in 2006. Sprint qualifying weekends, introduced in 2021 at selected events, further complicate direct comparisons. The analysis treats all qualifying positions equivalently, which may introduce minor inconsistencies across format changes.

11. Conclusion

This study has provided a comprehensive, data-driven examination of the relationship between qualifying performance and race outcomes in modern Formula 1, covering 25 seasons from 2000 to 2024 and 7,134 classified race finishes.

The central finding is unambiguous: qualifying position is a strong predictor of race finishing position. A Spearman rank correlation of 0.785 and Pearson correlation of 0.782 indicate a robust positive relationship that has remained consistent across the modern era. The MAE of 3.06 positions confirms that drivers tend to remain within approximately three positions of their grid slot, underscoring the decisive importance of track position in a sport where overtaking is structurally difficult.

The podium probability analysis reveals perhaps the most practically significant finding: drivers starting from the front three grid positions finish on the podium in nearly 74% of races, compared to under 16% for midfield starters and under 2% for those starting eleventh or lower. This extreme probability gradient, confirmed as statistically significant by a chi-square test with an effectively zero p-value, provides compelling evidence that the value of front-row qualification extends far beyond a single lap of performance.

Era-wise analysis adds temporal nuance: the qualifying–race relationship was strongest during 2010–2019 (Spearman = 0.813), corresponding to sustained single-team dominance and stable technical regulations. The 2000–2009 and 2020–2024 eras show slightly lower correlations, consistent with greater competitive variability.

The interactive Streamlit dashboard developed alongside this analysis enables stakeholders to explore these patterns dynamically, making the findings accessible beyond the static outputs of the statistical analysis. Taken together, the evidence from this project validates what experienced Formula 1 observers have long argued qualitatively: securing a strong qualifying position is a fundamental strategic advantage that dramatically shapes the probability of race success.